



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2022

**HSC
Assessment
Task 1**

Mathematics Extension 2

General Instructions

- Reading Time – 2 minutes
- Working Time – 58 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Section II, show relevant mathematical reasoning and/or calculations

Total marks – 39

Section I – 6 marks (pages 2 – 4)

- Attempt Questions 1 – 6
- Allow about 8 minutes for this section

Section II – 33 marks (pages 5 – 7)

- Attempt Questions 7 – 10
- Allow about 50 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER

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Question	1 – 6	7	8	9	10	Total
Complex Numbers	/6	/9	/7	/9	/8	/39

Section I

6 marks

Attempt Questions 1-6

Allow about 8 minutes for this section

Use the multiple choice answer sheet for Questions 1-6.

1 What is $-1+i\sqrt{3}$ in exponential form?

A. $2e^{i\frac{\pi}{3}}$

B. $2e^{i\frac{2\pi}{3}}$

C. $2e^{i\frac{4\pi}{3}}$

D. $2e^{i\frac{5\pi}{3}}$

2 The equation $x^2 - 2ix + k = 0$ has a root of $-1+2i$. What is the value of k ?

A. $-1+2i$

B. $-1-2i$

C. 1

D. 5

3 If $z = \omega$ is a non-real solution of the equation $z^3 = 1$, which of the following expressions is equal to $(1+\omega-\omega^2)^7$?

A. 128ω

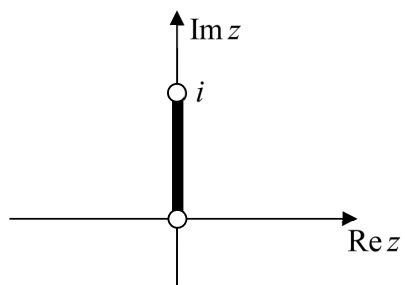
B. -128ω

C. $128\omega^2$

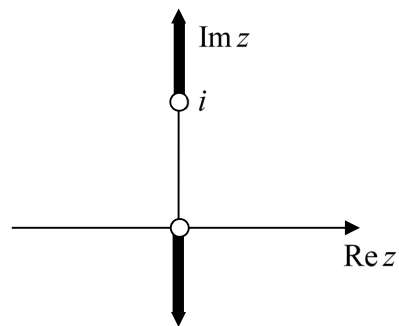
D. $-128\omega^2$

4 Which of the following loci is represented by the equation $\arg z = \arg i$?

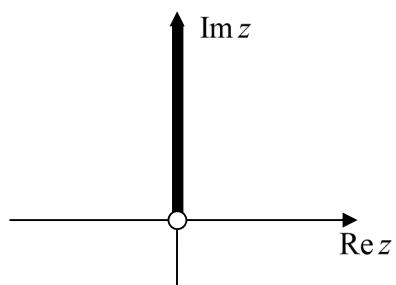
A.



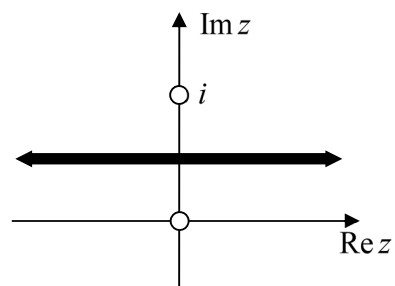
B.



C.



D.



5 Given $z_1 = 3 \operatorname{cis} \theta$ and $z_2 = 7 \operatorname{cis} \phi$, which of the following could be $|z_1 + z_2|$?

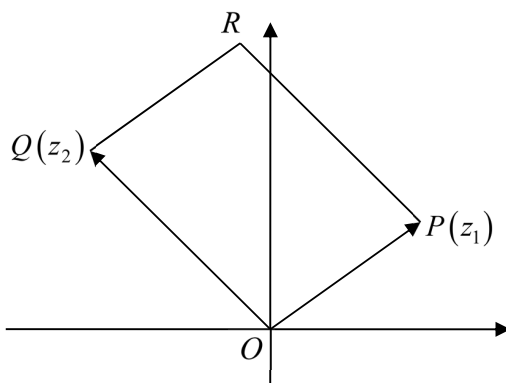
A. 1

B. 3

C. 7

D. 11

- 6 In the following diagram, $OPRQ$ is a parallelogram, and the vectors OP and OQ represent the complex numbers z_1 and z_2 respectively.



NOT TO SCALE

If $\frac{z_2}{z_1} = \sqrt{3}i$, what is the value of $\left| \frac{z_1 + z_2}{z_2 - z_1} \right|$?

- A. 1
- B. $\sqrt{3}$
- C. $\frac{1}{\sqrt{3}}$
- D. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Section II

33 marks

Attempt Questions 7-10

Allow about 50 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 7-10, your responses should include relevant mathematical reasoning and/or calculations.

Question 7 (9 marks)

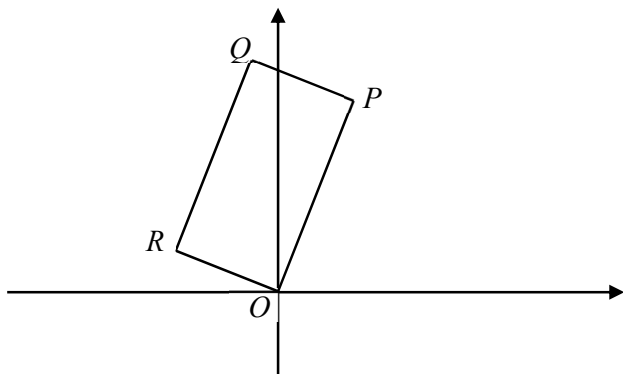
(a) Given $z = 2 - i$ and $w = 3 - 2i$, find in the form $a + ib$:

(i) $z + \bar{w}$ 1

(ii) $\frac{z}{w}$ 2

(b) Solve the equation $16iz^2 + (8 + 8i)z - 10 + 5i = 0$, expressing answers in the form $a + ib$. 4

(c) $OPQR$ is a rectangle and $OP = 3OR$. 2



If $\overrightarrow{OQ}$ represents the complex number $-1 + 6i$, determine what complex number is represented by the vector $\overrightarrow{OR}$.

Question 8 (7 marks) Start a new booklet.

- (a) Solve the equation $z^5 = 4 + 4i$. Give your answers in exponential form. **3**
- (b) The polynomial $P(x) = x^4 - 8x^3 + 24x^2 - 32x + 20$ has $1 + i$ as a zero.
- (i) Express $P(x)$ as a product of two real quadratic factors. **3**
- (ii) Explain briefly why $P(x) > 0$ for all real values of x . **1**

Question 9 (9 marks) Start a new booklet.

- (a) (i) On the same Argand diagram, sketch the locus of all complex numbers z which satisfy: **2**
 $|z - 3| = 1$ or $|z| = |z - 2i|$
- (ii) Hence write down the complex number(s) which satisfy: **1**
 $|z - 3| = 1$ and $|z| = |z - 2i|$
- (b) (i) Given that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for positive values of n , deduce that it is also true for negative values of n . **2**
- (ii) Hence show that if $Z = \cos \theta + i \sin \theta$ then $Z^n + Z^{-n} = 2 \cos n\theta$. **1**
- (iii) By considering $(Z + Z^{-1})^4$, show that $\cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3)$. **3**

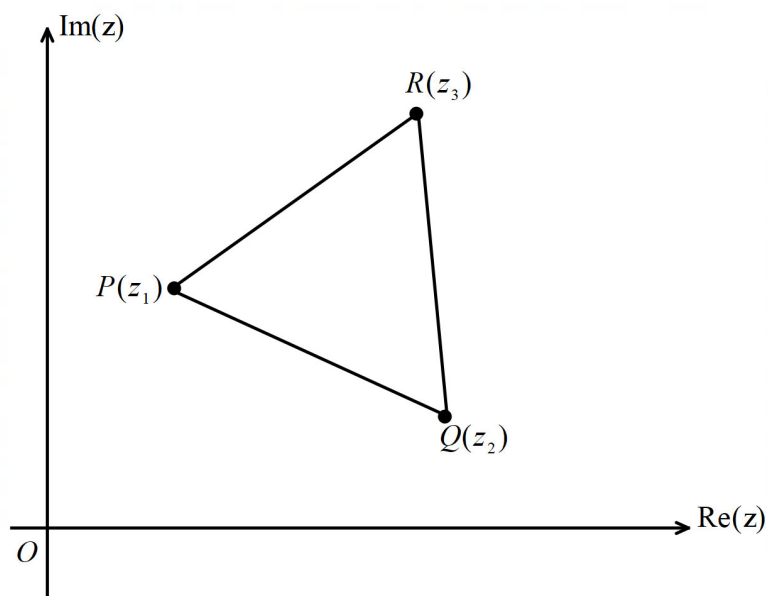
Question 10 (8 marks) Start a new booklet.

- (a) You are given that $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$. (DO NOT PROVE THIS) 3

Without using any other trigonometric identities, use this result to show that

$$\sin^3 x = \frac{3 \sin x - \sin 3x}{4}$$

- (b) z_1 , z_2 and z_3 are three complex numbers which are represented by the points P , Q , R respectively in the complex plane, where ΔPQR is equilateral.



- (i) Explain why $\frac{z_3 - z_1}{z_2 - z_1} = \operatorname{cis}\left(\frac{\pi}{3}\right)$. 1
- (ii) Hence or otherwise, show that $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_1z_3 + z_2z_3$. 2
- (iii) If z_1 , z_2 and z_3 are the three roots of the cubic equation $z^3 + az^2 + bz + c = 0$, find a relationship between a and b . 2

End of paper

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2022 Ext 2 Ass 1 Solutions

Section I

Summary: 1. B 2. A 3. D
4. C 5. C 6. A

$$\textcircled{1} |-1+i\sqrt{3}| = 2, \arg(-1+i\sqrt{3}) = \frac{2\pi}{3}$$

$$\therefore -1+i\sqrt{3} = \underline{2e^{i\frac{2\pi}{3}}}$$

$$\textcircled{2} (-1+2i)^2 - 2i(-1+2i) + k = 0$$

$$1-4i-4+2i+4+k=0$$

$$\underline{k = -1+2i}$$

$$\begin{aligned}\textcircled{3} (1+\omega-\omega^2)^7 &= (1+\omega+\omega^2-2\omega^2)^7 \\ &= (0-2\omega^2)^7 \\ &= -128\omega^{14} \\ &= -128(\omega^3)^4 \cdot \omega^2 \\ &= -128(1)^4 \omega^2 \\ &= \underline{-128\omega^2}\end{aligned}$$

$$\textcircled{5} ||z_1| - |z_2|| < |z_1 + z_2| < |z_1| + |z_2|$$

$$|3-7| < |z_1 + z_2| < 3+7$$

$$4 < |z_1 + z_2| < 10$$

$$\therefore |z_1 + z_2| = 7$$

$$\textcircled{4} \arg z = \arg i \\ = \underline{\frac{\pi}{2}}$$

$$\textcircled{6} \frac{z_2}{z_1} = \sqrt{3}i \Rightarrow |z_2| = \sqrt{3} \cdot |z_1|$$

and $OQ \parallel OP \Rightarrow OPRQ$ is a rectangle

$$\therefore |z_1 + z_2| = |z_1 - z_2| \quad (\text{diagonals of rectangle equal})$$

$$\therefore \left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$$

Question 7

$$(a) (i) \quad z + \bar{w} = (2-i) + (3+2i) \\ = \boxed{5+i} \quad (1)$$

$$(ii) \quad \frac{z}{w} = \frac{2-i}{3-2i} \times \frac{3+2i}{3+2i} \\ = \frac{6+4i-3i+2}{9+4} \\ = \boxed{\frac{8}{13} + \frac{1}{13}i} \quad (2)$$

(b)

$$16iz^2 + (8+8i)z - 10+5i = 0 \\ z = \frac{-(8+8i) \pm \sqrt{(8+8i)^2 - 64i(-10+5i)}}{32i} \\ = \frac{-8-8i \pm \sqrt{320 + 768i}}{32i} \\ = \frac{-1-i \pm \sqrt{5+12i}}{4i}$$

$$\text{let } \sqrt{5+12i} = a+ib, \quad a, b \text{ real} \\ 5+12i = (a^2-b^2) + 2abi$$

equating real & imag. parts:

$$a^2 - b^2 = 5 \quad (1)$$

$$2ab = 12 \Rightarrow b = \frac{6}{a} \quad (2)$$

$$\text{sub (2) in (1): } a^2 - \frac{36}{a^2} = 5$$

$$a^4 - 5a^2 - 36 = 0$$

$$(a^2-9)(a^2+4) = 0$$

$$a = \pm 3$$

$$b = \pm 2$$

$$\therefore \sqrt{5+12i} = \pm (3+2i)$$

$$\begin{aligned}
 z &= \frac{-1-i \pm (3+2i)}{4i} \\
 &= \frac{2+i}{4i} \times \frac{-i}{-i}, \quad \frac{-4-3i}{4i} \times \frac{-i}{-i} \\
 &= \frac{1-2i}{4}, \quad \frac{-3+4i}{4} \\
 &= \boxed{\frac{1}{4} - \frac{1}{2}i, -\frac{3}{4} + i}
 \end{aligned}$$

OR

$$\begin{aligned}
 16iz^2 + (8+8i)z - 10 + 5i &= 0 \\
 \times -i: 16z^2 + (8-8i)z + 5 + 10i &= 0 \\
 16z^2 + (8-8i)z - 2i &= -5-12i \\
 16z^2 + (8-8i)z + (1-i)^2 &= i^2(3+2i)^2 \\
 [4z + (1-i)]^2 &= i^2(3+2i)^2 \\
 4z + (1-i) &= \pm i(3+2i)
 \end{aligned}$$

$$4z + (1-i) = \pm (-2+3i)$$

$$4z = -3+4i, 1-2i$$

$$z = \boxed{-\frac{3}{4} + i, \frac{1}{4} - \frac{1}{2}i}$$

④

$$\begin{aligned}
 (c) \quad \vec{OQ} &= \vec{OP} + \vec{OR} \\
 -1+6i &= -3i \cdot \vec{OR} + \vec{OR} \\
 &= (1-3i) \cdot \vec{OR}
 \end{aligned}$$

$$\vec{OR} = \frac{-1+6i}{1-3i} \times \frac{1+3i}{1+3i}$$

$$= \boxed{-\frac{19}{10} + \frac{3}{10}i}$$

②

Question 8

$$(a) \quad z^5 = 4 + 4i$$

$$= 4\sqrt{2} \operatorname{cis}\left(\frac{\pi}{4} + 2k\pi\right), \quad k \in \mathbb{Z}$$

$$= (\sqrt{2})^5 \operatorname{cis} \frac{8k+1}{4} \pi$$

$$z = \sqrt{2} \operatorname{cis} \frac{8k+1}{20} \pi$$

$$k = -2, -1, 0, 1, 2: \quad z = \sqrt{2} \operatorname{cis}\left(-\frac{3\pi}{4}\right), \sqrt{2} \operatorname{cis}\left(-\frac{7\pi}{20}\right), \sqrt{2} \operatorname{cis}\frac{\pi}{20}, \sqrt{2} \operatorname{cis}\frac{9\pi}{20}, \sqrt{2} \operatorname{cis}\frac{17\pi}{20}$$

$$z = \sqrt{2} e^{-i\frac{3\pi}{4}}, \sqrt{2} e^{-i\frac{7\pi}{20}}, \sqrt{2} e^{i\frac{\pi}{20}}, \sqrt{2} e^{i\frac{9\pi}{20}}, \sqrt{2} e^{i\frac{17\pi}{20}} \quad (3)$$

$$(b)(i) \quad P(x) = x^4 - 8x^3 + 24x^2 - 32x + 20 = 0$$

$$P(1+i) = 0 \quad (\text{given})$$

$$\therefore P(1-i) = 0 \quad (\text{non-real roots of real polynomials occur in conjugate pairs})$$

let other roots be α, β

$$\text{sum: } (1+i) + (1-i) + \alpha + \beta = 8 \Rightarrow 2 + \alpha + \beta = 8 \Rightarrow \alpha + \beta = 6$$

$$\text{product: } (1+i)(1-i)\alpha\beta = 20 \Rightarrow 2\alpha\beta = 20 \Rightarrow \alpha\beta = 10$$

$$\therefore P(x) = (x^2 - 2x + 2)(x^2 - 6x + 20) \quad (3)$$

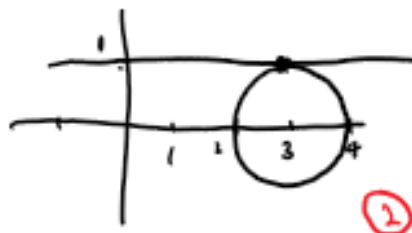
$$(ii) \quad x^2 - 2x + 2 = (x-1)^2 + 1 > 0 \quad \forall x \in \mathbb{R}$$

$$x^2 - 6x + 20 = (x-3)^2 + 11 > 0 \quad \forall x \in \mathbb{R}$$

$$\therefore P(x) > 0 \quad \forall x \in \mathbb{R} \quad (1)$$

Question 9

(a) (i)



(ii) $3+i$ (1)

(b) (i) Let $m = -n$ where $n > 0$, so $m < 0$

$$\begin{aligned} (\cos \theta + i \sin \theta)^m &= (\cos \theta + i \sin \theta)^{-n} \\ &= \frac{1}{(\cos \theta + i \sin \theta)^n} \\ &= \frac{1}{\cos n\theta + i \sin n\theta} \times \frac{\cos n\theta - i \sin n\theta}{\cos n\theta - i \sin n\theta} \\ &= \frac{\cos n\theta - i \sin n\theta}{\cos^2 n\theta + \sin^2 n\theta} \\ &= \frac{\cos(-n\theta) + i \sin(-n\theta)}{1} \\ &= \cos m\theta + i \sin m\theta \end{aligned} \quad (2)$$

$$\begin{aligned} (ii) \quad z^n + z^{-n} &= (\cos n\theta + i \sin n\theta) + (\cos(-n\theta) + i \sin(-n\theta)) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned} \quad (1)$$

$$\begin{aligned} (iii) \quad z^4 + z^{-4} &= (z + z^{-1})^4 - 4(z^2 + z^{-2}) - 6 \\ &= (z + z^{-1})^4 - 4[(z + z^{-1})^2 - 2] - 6 \\ &= (z + z^{-1})^4 - 4(z + z^{-1})^2 + 2 \\ 2 \cos 4\theta &= (2 \cos \theta)^4 - 4(2 \cos \theta)^2 + 2 \\ &= 16 \cos^4 \theta - 16 \cos^2 \theta + 2 \\ 8 \cos^4 \theta &= \cos 4\theta + 8 \cos^2 \theta - 1 \\ &= \cos 4\theta + 8 \times \frac{1}{2} (1 + \cos 2\theta) - 1 \end{aligned} \quad \begin{aligned} &= \cos 4\theta + 4 \cos 2\theta + 3 \\ \cos^4 \theta &= \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3) \end{aligned} \quad (3)$$

$$\begin{aligned}(z' + z^{-1})^4 &= (2\cos\theta)^4 \\ &= 16\cos^4\theta\end{aligned}$$

①

Also

$$\begin{aligned}(z' + z^{-1})^4 &= z^4 + 4z^3z^{-1} + 6z^2z^{-2} + 4zz^{-3} + z^{-4} \\ &= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6 \\ &= 2\cos 4\theta + 4(2\cos 2\theta) + 6 \\ &= 2\cos 4\theta + 8\cos 2\theta + 6.\end{aligned}$$

②

Equating expressions ① - ②

$$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$$

$$= 2[\cos 4\theta + 4\cos 2\theta + 3]$$

$$\cos^4\theta = \frac{1}{8}[\cos 4\theta + 4\cos 2\theta + 3]$$

Question 10

$$(a) \sin \theta = \frac{e^{-i\theta} - e^{i\theta}}{2i}$$

$$\sin^3 x = \frac{(e^{ix} - e^{-ix})^3}{8i^3}$$

$$= \frac{e^{3ix} - 3e^{ix} + 3e^{-ix} - e^{-3ix}}{-8i}$$

$$= -\frac{1}{4} \cdot \frac{e^{3ix} - e^{-3ix}}{2i} + \frac{3}{4} \cdot \frac{e^{ix} - e^{-ix}}{2i}$$

$$= -\frac{1}{4} \cdot \sin 3x + \frac{3}{4} \sin x$$

$$= \frac{3\sin x - \sin 3x}{4} \quad (3)$$

$$(b) (ii) \frac{z_3 - z_1}{z_2 - z_1} = \text{cis}\left(\pm \frac{\pi}{3}\right)$$

$$= \frac{1 \pm \sqrt{3}i}{2}$$

$$2(z_3 - z_1) = (1 \pm \sqrt{3}i)(z_2 - z_1)$$

$$= (z_2 - z_1) \pm \sqrt{3}i(z_2 - z_1)$$

$$2z_3 - z_1 - z_2 = \pm \sqrt{3}i(z_2 - z_1)$$

$$(2z_3 - z_1 - z_2)^2 = -3(z_2 - z_1)^2$$

$$4z_3^2 + z_1^2 + z_2^2 - 4z_1z_3 - 4z_2z_3 + 2z_1z_2 = -3z_2^2 - 3z_1^2 + 6z_1z_2$$

$$4z_3^2 + 4z_1^2 + 4z_2^2 = 4z_1z_2 + 4z_1z_3 + 4z_2z_3$$

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_1z_3 + z_2z_3 \quad (2)$$

(i) $z_3 - z_1$ & $z_2 - z_1$ have same length
so $\left| \frac{z_3 - z_1}{z_2 - z_1} \right| = 1$

and you rotate $z_2 - z_1$ 60° anti-clockwise
to get $z_3 - z_1$

$$\text{so } \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \frac{\pi}{3} \quad (1)$$

OR

$$z_3 - z_1 = \operatorname{cis} \frac{\pi}{3} (z_2 - z_1) \Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \frac{\pi}{3}$$

Similarly

$$z_3 - z_2 = \operatorname{cis} \frac{\pi}{3} (z_3 - z_1) \Rightarrow \frac{z_3 - z_2}{z_3 - z_1} = \frac{\pi}{3}$$

$$\therefore \frac{z_3 - z_1}{z_2 - z_1} = \frac{z_3 - z_2}{z_3 - z_1}$$

$$(z_3 - z_1)^2 = (z_3 - z_2)(z_2 - z_1)$$

$$z_3^2 - 2z_3z_1 + z_1^2 = z_3z_2 - z_2^2 - z_3z_1 + z_2z_1$$

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

$$(11) \quad z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_1z_3 + z_2z_3$$

$$(z_1 + z_2 + z_3)^2 - 2(z_1z_2 + z_1z_3 + z_2z_3) = z_1z_2 + z_1z_3 + z_2z_3$$

$$(z_1 + z_2 + z_3)^2 = 3(z_1z_2 + z_1z_3 + z_2z_3)$$

$$(\text{sum of roots})^2 = 3(\text{sum of roots in pairs})$$

$$(-a)^2 = 3b$$

$$a^2 = 3b$$

3